

Report
Of
Topic

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By

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CANDIDATE'S DECLARATION

I, Avishek Yadav do hereby declare that the internship report titled “**Topic**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

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4. Project Description

4.1 Introduction

Data analytics is the process of examining, cleaning, transforming, and interpreting data to discover useful information and support decision-making. The internship programme provided a structured progression from fundamental data handling to more advanced analytical methods, focusing on a practical component of the analytics workflow. It encouraged the preparation of professional deliverables such as notebooks, SQL files, visualizations, dashboards, reports, and automation scripts.

4.2 Organization Profile

CODTECH Private Limited focuses on digital solutions and technology services. The 30-day Data Analytics Internship was designed to provide hands-on learning opportunities and exposure to practical workflows, including data exploration, SQL, visualization, statistical analysis, and automation.

4.3 Problem Statement

Raw datasets commonly present challenges such as missing or inconsistent values, duplicate records, and the necessity to select suitable visualizations and analytical methods. The core problem addressed during the internship was converting this raw data into meaningful information through systematic cleaning, analysis, visualization, and interpretation.

4.4 Project Objectives

- To understand the complete data analytics workflow and set up a Python-based analytical environment.
- To clean datasets by handling missing values, duplicates, inconsistent data types, and potential outliers.
- To perform exploratory data analysis and use SQL for data extraction, aggregation, filtering, and joins.
- To create static and interactive visualizations and develop dashboards for presenting key performance indicators.
- To apply basic statistical analysis and prepare a final report emphasizing automation practices.

4.5 Scope of the Project

The scope encompassed a 30-day practical learning program spanning foundational setup, exploratory data analysis, SQL extraction, dashboarding, and advanced

predictive/segmentation techniques. Future scope extends to handling larger datasets, cloud databases, real-time dashboards, machine learning, and automated data pipelines.

4.6 Technologies and Tools Used

- Python, Jupyter Notebook, Pandas, and NumPy for data handling.
- Matplotlib, Seaborn, and Plotly for static and interactive visualization.
- SQL, SQLite/PostgreSQL concepts, and SQLAlchemy for Python–database integration.
- Power BI Desktop or Tableau Public for dashboarding.
- SciPy and Scikit-learn concepts for statistical analysis, clustering, and predictive modelling.
- GitHub and Microsoft Excel.

4.7 System Architecture

The data architecture followed a structured analytical workflow:

1. **Problem Understanding & Setup:** Defining objectives and preparing programming tools.
2. **Data Loading & Cleaning:** Importing datasets and handling missing values, duplicates, and incorrect data types.
3. **Exploratory Data Analysis:** Utilizing descriptive statistics and visualizations to understand distributions.
4. **Extraction & Visualization:** Using SQL for retrieval, followed by charting and dashboard creation.
5. **Advanced Analysis & Automation:** Applying statistical models, summarizing findings, and documenting repeatable code.

4.8 Methodology

The methodology involved a task-based progression. Task 1 focused on data quality assessment, using the IQR method for outliers and generating correlation heatmaps. Task 2 utilized SQL commands (SELECT, GROUP BY, JOIN) and Common Table Expressions to filter and aggregate structured data. Task 3 centered on visual storytelling via executive dashboards combining KPI cards and trend analysis. Task 4 introduced K-Means clustering, t-tests, chi-square tests, and train-test splitting for predictive modelling.

4.9 Expected Outcomes

- **Technical Proficiency:** Enhanced skills in data cleaning, EDA, Python, SQL querying, and database integration.
- **Analytical Reasoning:** The ability to move beyond descriptive reporting into hypothesis testing and customer segmentation.
- **Professional Deliverables:** Creation of reproducible workflows, professional presentations, and executive-style reports.
- **Reliable Insights:** A practical understanding that reliable insights depend on well-prepared data and clear communication.

4.10 Certificates of completion and other communication mail proof must be attached

(Attached in physical/final document as per institutional guidelines)

5. Bibliography/References

1. CODTECH Private Limited – 30-Days Data Analytics Internship Program material.
2. Internship task structure and prescribed activities provided for the programme.
3. Standard documentation and learning resources associated with Python, SQL, visualization, dashboarding, and introductory analytics tools.